7

Practical aspects of infra-red remote thermometry

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7.1 Introduction

All objects emit electromagnetic radiation, the quantity of emitted energy and its distribution in wavelength are strong functions of temperature. The filament of a light bulb emits a great deal of radiation in the visible part of the spectrum. To achieve this emission of light, the temperature of the filament is typically raised to well over 2000ºC. A household heating radiator is normally operated at around 70ºC, but our skin can detect the heat from it even without touching. The energy we feel is infra-red radiation.

Infra-red thermometers are devices that collect and detect infra-red radiation and deduce a temperature for the source. The inherent feature of infra-red thermometry is measurement at a distance – non-contact, non-invasive and non-contaminating. They are widely used throughout process industry, particularly where the measurement of a moving solid material, such as steel in a rolling mill or meat pies on a conveyor, is needed. Infra-red thermometers come in two fundamental forms, portable 'hand held' and installed 'fixed on-line' systems. The hand-held devices span from low cost 'guns', which measure a single spot, to highly advanced thermal cameras, capable of mapping the temperature of the target object. Installed 'fixed systems' also include both single-spot and imaging types, and in addition line-scanning types where a single spot is repeatedly scanned along a line over the surface of a moving object to produce profiles or once again a two-dimensional mapping.

There is a misconception that infra-red thermometry is applicable only at high temperatures; in reality, with modern equipment, measurements below 0ºC are quite common. In many circumstances the measurement can be very accurate indeed. Fibre-optics cables can in some situations be used to help access products in hostile conditions and measurements inside closed vessels are possible with suitable infra-red transmitting windows. The use of infra-red technology within the food industry has so far been limited, the reasons usually cited are:

- high costs when compared to traditional insertion probe equipment
- a perceived need to measure core or centre temperatures
- a lack of really suitable equipment available.

All three comments have some justification, but food technologists, like the metallurgists of the steel industry before them, will discover that a continuous measurement of surface temperature is much more useful for process control than an intermittent, manual, centre reading, and new instrument developments are improving the suitability situation.

Certain developments within the food industry provide strong incentives for transition to infra-red methods. Specific examples are microwave heating, which has problems associated with the use of conventional thermometers, and the rapid increase in chilled prepared foods, with the associated need for rapid, non-contaminating temperature checks. More particularly, movement from batch production methods to continuous processing favours non-contact techniques which can provide continuous on-line temperature monitoring and control. There is little published on the practicalities of infra-red thermometry, particularly at lower temperatures. It is hoped that this chapter goes some way to filling that gap.

7.2 Radiation thermometers

A radiation thermometer (Fig. 7.1) comprises three essential elements:

- an infra-red energy collection system
- an infra-red detector and dedicated precision electronics
- a display usually with some electronic signal processing.

Other topics in this section are housings and configurations, linescanning and thermal imaging cameras.

7.2.1 Collection

The infra-red wavelengths of interest in radiation thermometry (1 to 15 μm) are not so very much longer than the wavelengths of visible light (0.4 to 0.7 μm). As a result, infra-red thermometers look like optical instruments, rather than the plumbing or antennae of microwave or radio frequency equipment, and they operate according to simple laws of ray optics.

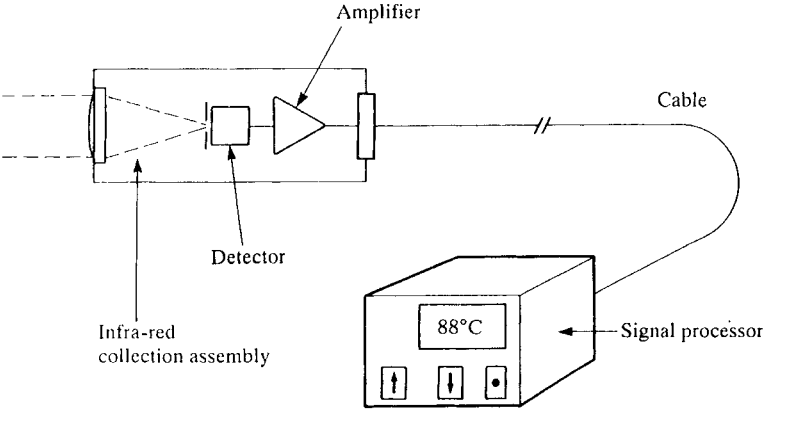


Fig. 7.1 Schematic diagram of an infra-red thermometer.

A well designed instrument collects radiation only from within a precisely defined field of view; Fig. 7.2 shows a typical example. In this example the field is axially symmetrical and necks down from 25 mm diameter at the lens of the thermometer to 10 mm at the best focus point, diverging thereafter. The object to be measured need not be at this best focus point, and a thermometer will measure any object which intersects this cone of vision.

In the example shown, an object just 400 mm in front of the thermometer would need to be larger than the 19 mm diameter of the cone for reliable measurements. If the object to be measured does not completely fill the cone, then some radiation will be collected from other items behind the intended target, and the thermometer will measure a weighted average of all those objects in view. Thus, for a well-designed instrument the measurement is independent of object distance. In particular, the focus point has no significance to the user beyond being the point at which the target spot is at its smallest, in relation to distance. The measurement is also insensitive to obliquity of the target surface, typically for angles up to 50° off normal.

We all know glass is transparent to visible light, so the use of a thermometer collecting energy near the visible spectrum to measure a glass object would be fruitless. In this case the instrument would look straight through the glass and measure whatever was behind the glass object. Fortunately, the vast majority of materials are highly opaque in the infra-red part of the spectrum. The emitted energy originates only just beneath the target surface. Penetration depths are typically reckoned in micrometres and the temperature measured is, to all intents and purposes, a surface temperature. Many instruments have adjustable focus. Typically, the angular field of view is preserved, at least approximately, while the focus distance changes. For example, the instrument of Fig. 7.2 might be adjustable to a 5 mm target spot at a distance of 500 mm or a 30 mm spot at 3000 mm.

Most instruments incorporate some form of aiming system to allow the user to identify the target being measured. Typically this takes two forms:

- Visual sighting, in which a through-the-lens optical arrangement allows the user to view the target through an eyepiece. A circular graticule built into the eyepiece provides a visual definition of the actual target being measured.
- Laser pointers, in which a low-power laser beam is projected from the thermometer onto the target surface. In some cases this is a single beam which simply illuminates the centre point of the target. More usefully, other designs have the laser beam scanned to draw a circular outline of the cone of view onto the target.

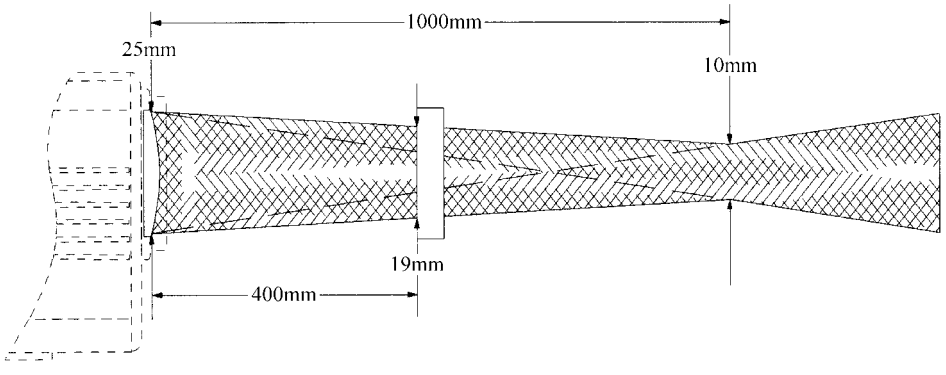


Fig. 7.2 Schematic diagram of thermometer field of view.

Where access to the object to be measured is restricted the collection optics can be decoupled from the detector by a fibre optics light guide. This allows smaller optical systems, which are capable of withstanding more extreme environments.

7.2.2 Detectors

The detector transduces the collected infra-red energy to an electrical signal. Typically, this signal is very small and the detector must be combined with precision electronics to achieve a more usable level. The detector usually incorporates a spectral filter that defines the range of infra-red wavelengths sensed by the thermometer. We shall see in the next section that the operating waveband has a critical influence on instrument accuracy and practical performance.

Details of detector operation need not concern the instrument user, but some appreciation of the available types and their performance limitations is valuable. Table 7.1 lists the operating waveband and detector combinations most commonly used in industrial general-purpose thermometers, together with the target temperature ranges over which they are most useful. Note that the emitted radiation both increases rapidly and shifts to shorter wavelengths as temperature increases.

An infra-red thermometer must inevitably view a target through the intervening atmosphere, so the infra-red transmission of this atmosphere must be considered. Many of the wavebands listed in Table 7.1, correspond to atmospheric transmission 'windows' (Fig. 7.3), i.e. regions of the infra-red spectrum where the atmosphere is highly transparent. Atmospheric absorption primarily derives from carbon dioxide and water vapour; symmetrical molecules (O_2, N_2, etc.) have no absorption bands in the infra-red region. A thermometer with a poorly chosen waveband will give readings that depend on target distance and also on humidity. The potentially very useful 3 to 5 μm window is widely used in linescanners and thermal imagers (see Sections 7.2.5–7.2.6), and recent developments have seen the adoption of this waveband for spot thermometers with capability to measure down to 0ºC.

The silicon photovoltaic diode is a superb detector of infra-red near 1 μm wavelength. The detection mechanism is electron-hole pair generation by photons absorbed in a p–n junction. The signal current is linear with incident energy to better than 1 per cent over

Table 7.1 Operating wavebands and detector types for commonly available infra-red thermometer types

Operating waveband (μm)	Detector types	Most useful range (ºC)
0.7 to 1.0	Silicon	400+
1 to 1.7	InGaAs Germanium	200+
1.8 to 2.5	InGaAs Lead Sulphide	50 to 600
3 to 5	Lead Selenide	0 to 500
8 to 14	Thermopile Pyroelectric	−50 to 300

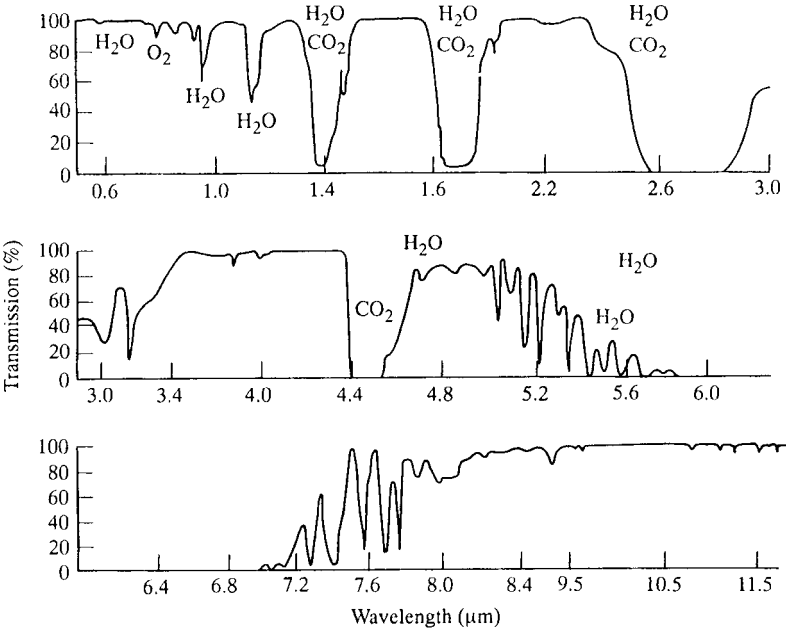


Fig. 7.3 Spectral transmission of the atmosphere: 300 m path length at sea level with 5.7 mm precipitable water at 26°C (Wolfe and Zissis 1978).

four to five decades. With careful amplifier design, drift is negligible. The effective gain varies with ambient temperature, but in a predictable manner which can be accurately compensated. There are no significant ageing mechanisms and fast (microsecond) response times are achievable. Silicon's usefulness is limited to target temperatures above about 400°C; the amount of 1 μm energy available for detection falls away very rapidly at lower temperatures. Efforts to squeeze lower temperature measurements by use of elaborate amplification schemes run into problems with reflected daylight.

Germanium operates in a similar fashion to silicon, but has a smaller bandgap, allowing absorption of lower energy, longer wavelength photons. It has all of silicon's attributes provided it is kept reasonably cool. Industrial germanium thermometers are typically specified to 50°C ambient but, in practice, benefit from the use of a water cooling jacket. Germanium is useful for target temperatures above about 200°C.

Developments in semiconductor materials have yielded a new type of infra-red detector based on an alloy of indium, gallium and arsenic (InGaAs). These are again photovoltaic diodes with a bandgap slightly wider than germanium, yielding detection of photons at slightly shorter wavelengths. These detectors are somewhat more tolerant to high ambient temperatures than germanium and offer excellent linearity over many decades of signal range. Derivatives of these InGaAs detectors are now available where the bandgap has been stretched by introducing other materials to the alloy, producing sensitivity at wavelengths as long as 2.4 μm.

Lead sulphide is a photoconductor, the detection mechanism being photon-induced resistance change in the bulk semiconductor. It is very sensitive but unstable. It is a practical necessity to modulate the incoming infra-red radiation and amplify the resultant a.c. signal. Modulation is invariably accomplished with some form of mechanical chopper, introducing issues of instrument reliability and service life. The detector sensitivity is subject to ageing and hysteresis effects, so good handling practices are

required during assembly and calibration. Good thermometer designs incorporate lead sulphide only as a sensitive element and not as a calibrated device. Regrettably, many instruments have been marketed which ignore this principle; they exhibit serious calibration drifts and have a tendency to 'remember' past illumination. Lead sulphide is useful down to target temperatures of around 50ºC. With good design it is the basis of the most accurate thermometers at these temperatures.

Lead selenide is another type of photoconductor and has very similar operating characteristics to the lead sulphide detector described above. The high sensitivity of lead selenide extends to a longer range of wavelengths allowing the minimum measured temperature to reach down to below 0ºC.

Thermopiles and pyroelectrics are thermal detectors whose sensitivity extends over all wavelengths. In general, the sensitivity of thermal detectors is significantly lower than that of the photon detectors described above. Thermopiles are easy to use and form the basis of a wide variety of simple and generally trouble-free thermometers. The response times are typically limited to about 50 ms. Pyroelectric detectors are somewhat more sensitive and faster (a few milliseconds) but a chopped radiation input is fundamental to the detection mechanism. Pyroelectrics are common in portable (i.e. intermittent use) thermometers, but the absence of moving parts is a powerful argument in favour of thermopiles for continuously used installed systems. Used with an 8 to 14 μm filter, thermal detectors allow measurement down to room temperature and below.

7.2.3 Processing electronics

The signals generated by the detectors are both non-linearly related to temperature and subject to the influence of target emissivity. The prime function of any processing electronics is to generate a temperature value by performing an algorithm of the general form:

$$T = f(\epsilon, T_i, S)$$

Where the inputs ϵ, T_i, S are, respectively, a user-set 'emissivity' factor, the thermometer internal temperature, and the detector signal. The target temperature T is typically output as an analogue signal (e.g. 4 to 20 mA) plus a digitally displayed value. The fast response of most infra-red thermometers allows the measurement of fleetingly brief intermittent events, such a objects falling through the view of the thermometer. The second major function of the processing electronics is to extract usable control parameters from the raw measured values, using real-time signal processing.

Microprocessor-based electronics conventionally take the form of a panel-mounted unit, remote from the sensor, with digital temperature display and user keypad. These units are very flexible, and many processing functions may be available beyond the basic derivation and display of target temperature. For example, the user may be able to select peak or valley picking of the output signal with adjustable decay times; sample/hold actuated by an external trigger signal; high/low alarm activation with adjustable limits; on/off, PID (proportional integral differential) or more complex forms of control with adjustable set-point and other parameters. The microprocessors range from 8 to 32 bit single-chip microcomputers with the program stored in EPROM and keypad set parameters stored in NOVRAM. A very high level of noise immunity is an essential feature for industrial use, particularly in heavy plant, requiring careful attention to both hardware design and software self-checking.

Analogue processors continue to play an important role and there are good reasons for this in terms of both performance, and cost-effectiveness. Digital processors are often more expensive, power hungry, and too slow for some applications. It should be noted that, at short wavelengths, analogue to digital conversions with dynamic ranges equivalent to 20+ bits, are typically needed to accommodate the extreme non-linear radiance/temperature relation. The best of the analogue systems display elegance bordering on genius and accommodate, with just a few components, enormous dynamic ranges in radiance with response times down to a few microseconds, or power requirements down to a few hundred microwatts. Virtually drift-free performance is achievable with good design. Analogue processing has retained a firm place for 'in sensor' processing. However, it cannot match microprocessors for the ever increasing need for flexibility and today digital processing is more frequently found embedded within the sensor.

7.2.4 Housings and configurations

As stated earlier, infra-red thermometers come in two basic forms:

- portable 'hand held' devices including low cost 'guns', high-quality thermometers with 'through the lens' sighting and thermal imagers, capable of temperature mapping of the target object.
- installed 'fixed systems' also including single spot and imaging types, and in addition linescanning types.

Figure 7.4 shows a high-specification, portable radiation thermometer (Cyclops 33). This instrument covers the range −50 to +1000ºC. It has an 8 to 14 μm operating waveband and uses a pyroelectric detector. The optics define a one degree field of view, with focus adjustable from 750 mm to infinity. Through the lens visual sighting is included. The measured temperature is displayed in the viewfinder and is also available as a digital output. Power is from a single PP3 9V battery. A host of accessories are available, including a data logger, printer and D/A converter.

The instrument shown is one of a range offering various temperature spans and operating wavelengths within similar mechanical configurations. This range actually derives from a manufacturing agreement between a company specialising in infra-red thermometry (Land) and a household-name camera manufacturer (Minolta). The success of this arrangement has, in recent years, spawned several such collaborations and



Fig. 7.4 High-specification portable radiation thermometer Cyclops 33. (Courtesy of Land Infrared Ltd).



Fig. 7.5 Low cost portable infra-red thermometers.

something of a revolution in the cost and quality of portable radiation thermometers. Figure 7.5 shows low-cost portable instruments (Mini Laser and Mini View) which are attracting interest in the food industry. They offer many of the features of the first instrument at a fraction of the cost.

Fixed instruments can be broadly divided into stand-alone and thermometer plus processor configurations. Figure 7.6 shows a family of thermometers and processors that can be combined to give the functions and features needed. The optical collection, detector and associated amplification are housed in a precision die-cast metal body which is connected, via a cable, to a remote processor and display unit. The sensors shown include a normal (focusable) optical type with through the lens sighting and a fibre optics version. These are both from the same family (System 4) which includes various wavebands and temperature spans in a common format. These sensors may be connected without any re-calibration to any one of a family of processors offering different levels of features and performance. For example, one model (LMG) provides graphical display of temperature to show trends and variances.

Figure 7.7 shows a stand-alone thermometer (SOLO 3). Here the collection, detection and processing are all contained within a single housing. The instrument shown is an 8 to 14 μm thermopile type measuring 0 to 250ºC. Again it is one of a range. Stand-alone instruments are typically available in 'two-wire' and 'four-wire' configurations. In four-wire types there are separate connections for power-input and for signal output. In the



Fig. 7.6 Family of interchangeable sensors and processors.

increasingly popular two-wire loop arrangement there are just two connections and the signal current is superimposed on the current drawn from the power supply. Usually it is arranged for the loop current to be 4 mA at the bottom of the measurement range and 20 mA at the top, with linear interpolation in between. Clearly, the whole of the thermometer's internal electronics must operate on less than 4 mA, and this results in performance limitations.

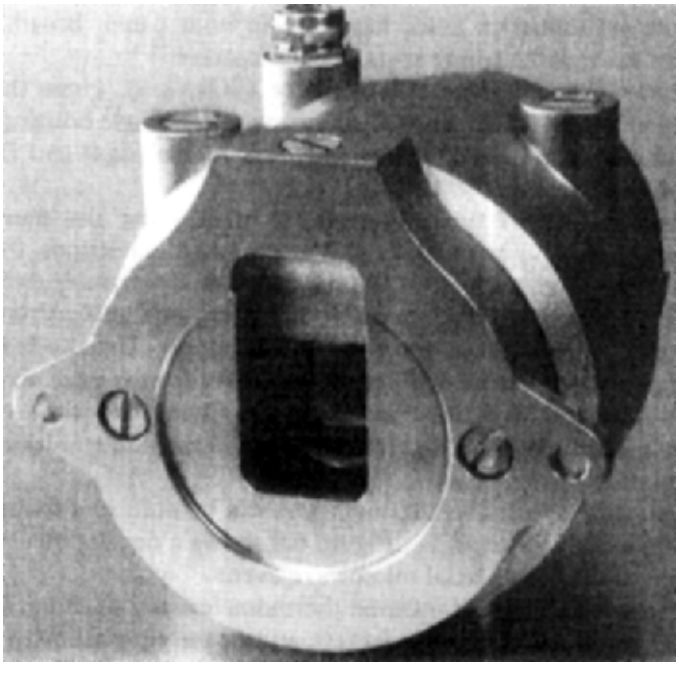


Fig. 7.7 Stand-alone bolt-on infra-red thermometer SOLO 3. (Courtesy of Land Infrared Ltd).

For special applications, infra-red thermometers can be built in a multitude of shapes and sizes. For example, Fig. 7.8 shows a developmental miniature unit for use in commercial microwave ovens. An essential part of any range of infra-red thermometers is a selection of mounting fixtures, water and air cooling jackets, and air purge attachments. The latter in particular have reached an advanced state of development and are capable of maintaining clean thermometer optics over very extended periods, even under the most appalling conditions in heavy industrial plants.

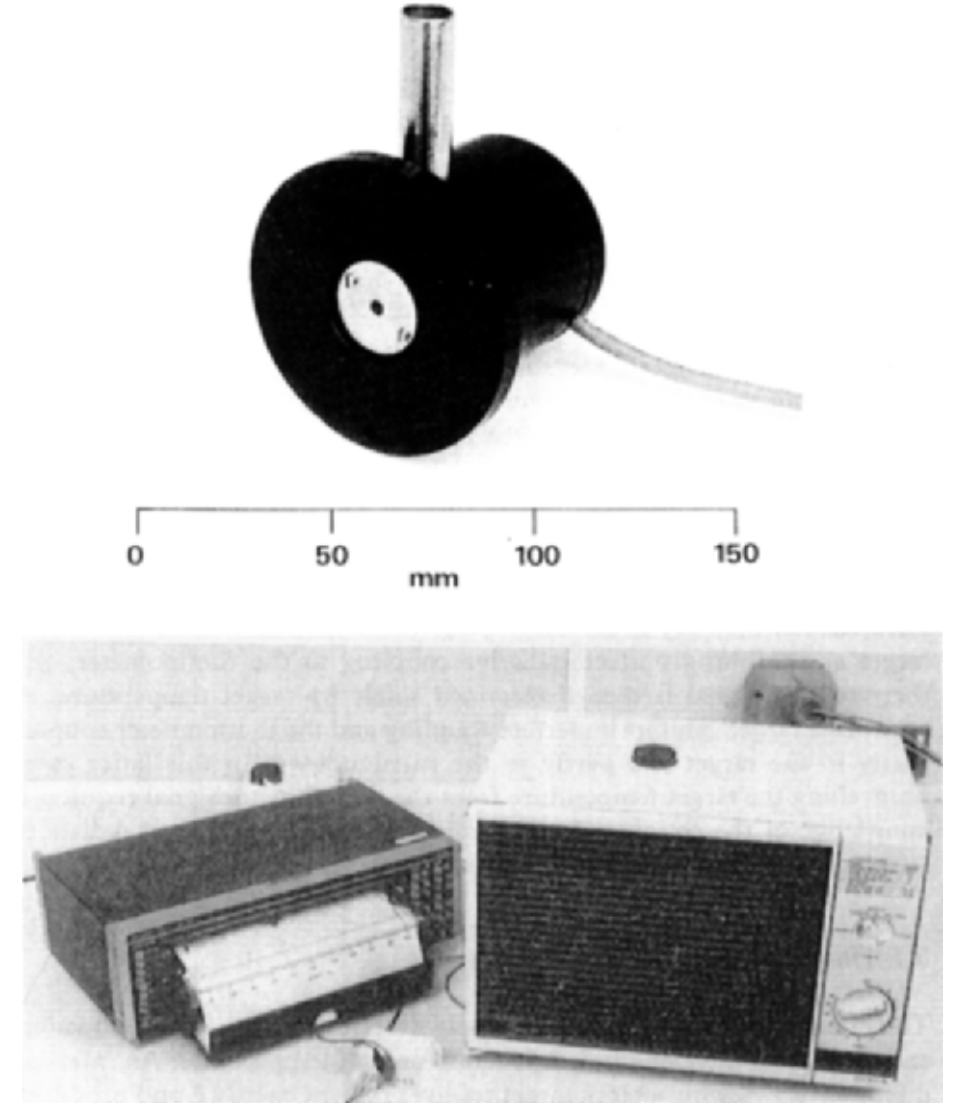


Fig. 7.8 Developmental miniature infra-red thermometer for microwave ovens, and same fitted to oven ceiling: the spigot allows purge air derived from the oven fan to enter the probe. (Courtesy of Land Infrared Ltd).

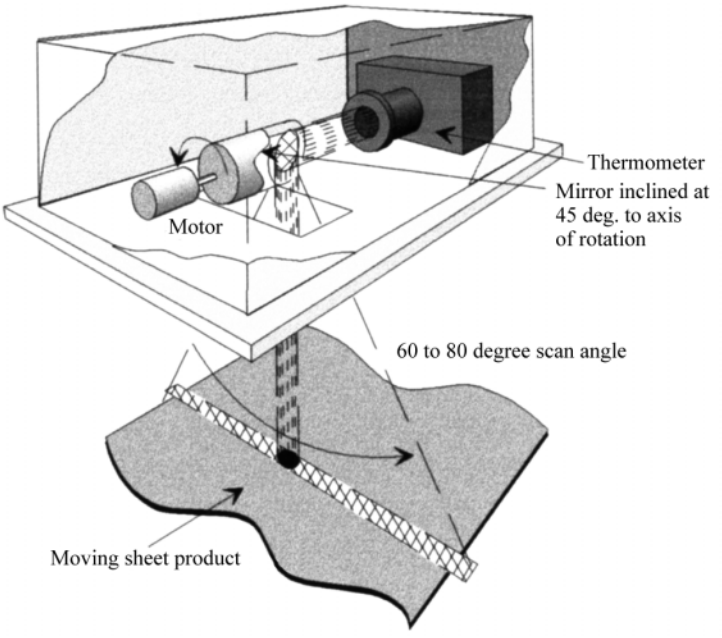


Fig. 7.9 Linescanning thermometer.

7.2.5 Linescanning thermometers

A linescanning thermometer measures a temperature profile across the target's surface. It is essentially an infra-red thermometer, whose field of view is scanned, in one dimension, by some rotating mirror arrangement, see Fig. 7.9. So far, few applications have called for a hand-held version and those available have very limited spatial resolution. Typically a scanner is used to produce a two-dimensional map of a continuous stream of product in sheet form, such as hot rolled steel sheet or extruded pastry on a conveyor. With scan repetition rates of up to one hundred lines per second and up to one thousand measured samples per scan line, large amounts of data are generated and in most cases the output signal is processed by a computer to produce a temperature/position trace. Highly elaborate data processing schemes are available, for example, allowing detection of differences between a stored 'ideal' profile and the live data from the scanner, or the determination of temperature time trends at prescribed positions across the scanned target.

Line scans measuring down to room temperature operate at 3 to 5 μm using thermoelectrically cooled photoconductors – lead selenide or mercury cadmium telluride. A typical specification might be 10 scans per second with a spatial resolution of one-hundredth of the separation of the instrument from the target. In terms of accuracy of temperature measurement, exactly the same factors apply as for a single-spot thermometer.

7.2.6 Thermal imaging cameras

In a thermal imager the field of view is scanned in a two-dimensional raster and each point in the image plane is handled in the same way as a thermometer target point. Image data is generated with the temperature of each point indicated on a density (grey) scale or



Fig. 7.10 Thermal imager for fixed installation.

false colour temperature contour map. Modern industrial thermal imagers operate at 3 to 5 μm using thermoelectrically cooled mercury cadmium telluride. Good quality (200+ lines) images are generated at twenty frames per second with minimum resolvable temperature difference a tenth of a degree Celsius. Integration of successive frames allows temperature differences below 0.1ºC to be resolved. Devices used in research and medicine and by the military achieve superior performance using detectors cooled with liquid nitrogen; both 3 to 5 μm and 8 to 14 μm wavebands are used.

Thermal imaging cameras are most commonly found in portable, hand-held forms, with the image data viewed directly by the user. Initially developed as military search tools, these devices have primarily found applications in maintenance and inspection roles, where hand-held or portable devices have been ideal. Recently, several applications have required the permanent installation of thermal imaging cameras linked to powerful image processing software, to provide continuous monitoring of things as diverse a bulk coal storage, or combustion chamber safety monitoring. Instruments specifically designed for continuous on-line use are now available (see Fig. 7.10).

Historically, thermal imager development has focused on spatial and temperature resolution, to the detriment of the absolute radiance calibration, which is a prerequisite for accurate temperature measurement. However, recent devices claim temperature accuracy comparable with that of single-spot thermometers of similar operating wavebands.

7.3 Measurement principles

An infra-red thermometer directly measures infra-red energy, specifically the radiance of the target. From this radiance measurement a target temperature is deduced, using a model, based on device characteristics, linking temperature and radiance. Formulation of a satisfactory model can be trivial or practically impossible, dependent on often quite

subtle aspects of the measurement situation. Herein lies much of the fascination of infra-red thermometry and the source of its reputation as something of a black art.

One should recognise that the need for an interpretative model is not unique to infra-red techniques. Consider a mercury-in-glass thermometer: if the 'target' is a stirred liquid then there is little problem in interpreting the measured height of the mercury column in relation to the liquid temperature. However, if the bulb is merely pressed against the outside of a plastic bottle of indifferent conductivity, perhaps in a draught and with some incident radiation, then concluding what, if anything, the thermometer is telling us about the bottle temperature less still the liquid inside, is far from simple.

The essence of contact thermometry is thermalisation; the allowance that the sensor must reach the same temperature as the object to be measured. For infra-red thermometry the black body radiator is the equivalent. One can view a 'black' target as exhibiting perfect radiative coupling to the thermometer; the thermometer signal is then determined solely by target temperature. Real-world non-black body targets exhibit imperfect coupling and the thermometer couples partly to the target and partly to the surroundings. In this latter case, unravelling the target temperature from the thermometer signal requires some knowledge of the coupling factor (emissivity) and background radiance (conveniently expressed as a mean background temperature).

7.3.1 The Planck function and black bodies

The basic law for thermal radiation was derived by the eminent physicist Max Planck, whose equation relating emitted radiance of a black body to wavelength and temperature, known universally as the Planck equation, is given below.

$$J_{\lambda T} = C_1 \cdot \lambda^{-5} \cdot \{\exp(C_2/\lambda T) - 1\}^{-1} \tag{1}$$

Figure 7.11 shows this function evaluated over a range of wavelengths and temperatures. It can clearly be seen that both the quantity and spectral distribution of the energy vary markedly with temperature. An infra-red thermometer viewing a black body target receives radiant energy L and produces a proportional detector signal, S:

$$S \propto L = f(T) \tag{2}$$

where the calibration function $f(T)$ is the Planck equation integrated over the thermometer operating waveband:

$$f(T) = \int_{\lambda_1}^{\lambda_2} J_{\lambda_T} \cdot S_\lambda \cdot d\lambda \tag{3}$$

If, for simplicity, we consider a thermometer with a narrow waveband $\delta\lambda$ at wavelength λ, then:

$$L \approx C_1 \cdot \lambda^{-5} \cdot \{\exp(C_2/\lambda T) - 1\}^{-1} \cdot \delta\lambda \tag{4}$$

where $C_1 = 1.191 \times 10^{-16}$ and $C_2 = 1.438 \times 10^{-2}$ in SI units, and T is in Kelvin.

The inherent sensitivity of any instrument is the rate of change of the measured parameter, in this case radiance L, with respect to the intended measured variable, here temperature, T. For infra-red thermometers this is often expressed as fractional change thus:

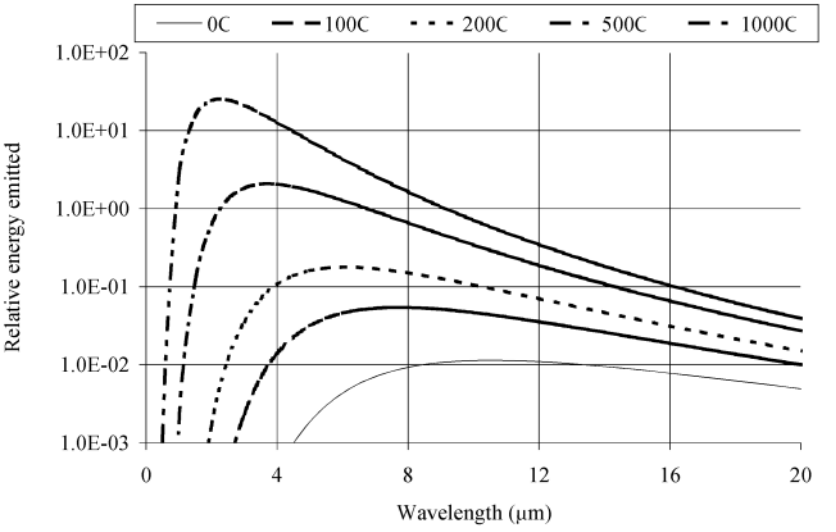


Fig. 7.11 Spectral distribution of infra-red energy.

$$\frac{\delta L}{\delta T} \cdot \frac{1}{L} = \exp(-C_2/\lambda T^2) \tag{5}$$

Table 7.2 gives $f(T)$ evaluated for narrow band thermometers operating at various wavelengths together with inherent sensitivity, expressed in per cent change in radiance L required to produce a one degree change in indicated target temperature. For any given temperature this per cent per degree (%/ºC) indicates much higher inherent sensitivity at shorter wavelengths.

7.3.2 Non-black targets and emissivity

The Planck function describes the emission from an idealised perfect surface, known as a black body. Laboratory targets are readily constructed which are almost perfectly black. Real objects to be measured emit only a fraction of that from a black body at the same temperature, and this fraction, known as emissivity, is commonly represented by the Greek letter ϵ. Emissivity can theoretically vary between unity for black body emitters and zero for perfect reflectors.

Many common materials – for example, anything containing enough water – are very close to black. Others, like aluminium, are very far from black, emitting only some 5 per cent of the energy of a black body. Care should be taken over the interpretation of the word black, as the colour of an object as seen by our eyes in the visible part of the spectrum often has little bearing on its emissivity in the infra-red. For a non-black target, equation (2) becomes:

$$S \propto L = \epsilon \cdot f(T) + (1 - \epsilon) \cdot f(T^{\sim}) \tag{6}$$

where the first term represents emission from the target and the second represents radiation originating in the surroundings and reflecting off the target spot into the thermometer, ϵ is the emissivity factor. $T^{\sim}$ is an effective background temperature, a formalism for describing a potentially very complex reflection situation. To resolve equation (5) it is necessary to ascribe values to ϵ and $T^{\sim}$.

Table 7.2 Calibration function $f(T)$ in arbitary units and rate of change of $F(T)$ for varous narrow-band thermometers.

Target temperature	1 μm		1.6 μm		2.2 μm		4 μm		10 μm	
(ºC)	*f(T)*	%/ºC	*f(T)*	%/ºC	*f(T)*	%/ºC	*f(T)*	%/ºC	*f(T)*	%/ºC
-20	–	–	–	–	–	–	–	–	4.05E+00	2.3
0	–	–	–	–	–	–	2.21E-01	4.9	6.16E+00	2.0
20	–	–	–	–	–	–	5.42E-01	4.3	8.84E+00	1.7
40	–	–	–	–	1.95E-03	6.9	1.19E-00	3.7	1.21E+01	1.5
60	–	–	–	–	6.83E-03	6.1	2.37E-00	3.3	1.60E+01	1.3
80	–	–	–	–	2.08E-02	5.4	4.37E-00	2.9	2.06E+01	1.2
100	–	–	–	–	5.61E-02	4.8	7.54E-00	2.6	2.57E+01	1.1
150	6.65E-03	5.1	4.46E-01	3.7	2.36E+01	2.0	4.11E+01	0.8		
200	–	–	6.29E-02	4.1	2.29E-00	3.0	5.80E+01	1.6	5.97E+01	0.7
300	1.48E-03	4.5	1.74E-00	2.8	2.55E+01	2.0	2.19E+02	1.1	1.05E+02	0.5
400	6.18E-02	3.2	1.79E+01	2.0	1.39E+02	1.5	5.58E+02	0.8	–	–
500	9.82E-01	2.4	1.01E+02	1.5	4.89E+02	1.1	1.12E+03	0.6	–	–
600	8.28E-00	1.9	3.82E+02	1.2	1.29E+03	0.9	–	–	–	–
800	1.79E+02	1.3	2.60E+03	0.8	–	–	–	–	–	–
1000	1.47E+03	0.9	9.72E+03	0.6	–	–	–	–	–	–

Most thermometers incorporate a means to input an emissivity value based on prior knowledge. Emissivity is primarily a function of target material and surface condition, but also depends on wavelength and, to a small extent, on temperature. In high-accuracy thermometry it must be regarded as an empirically determined parameter and most reputable suppliers of infra-red thermometers offer emissivity determination as a service at modest cost. Given the repetitive nature of modern manufacturing processes and the (usual) emphasis of measurement repeatability over absolute accuracy, it is the effect of fluctuations in emissivity that need to be carefully assessed.

Various tabulations of emissivity values exist, but need to be treated with great care. Sometimes the values quoted are wide-band, i.e. integrated over all wavelengths, appropriate to heat transfer calculations but useless for thermometry. Sometimes values relate to laboratory-pure specimens which bear little resemblance to target surfaces found in the real world.

Some generic examples appropriate to a wavelength of 10μm and low temperatures are as follows. Water is extremely black near 10μm, and anything with a wet surface film – slurries, pastes, flesh – may be expected to have an emissivity approaching 0.99. Many fat-based products – e.g. chocolate – have high emissivities of about 0.85. A vast number of organic or fibrous materials have emissivities of around 0.8. Most paints, varnishes, etc. have an emissivity of about 0.8, irrespective of visible coloration. Opaque plastics typically have an emissivity of 0.96; rubber 0.9; glass 0.85; wood 0.85; asbestos 0.9; most building materials, i.e. brick, cement etc., 0.7–0.9; paper and cardboard 0.8–0.95; cloth 0.75. Black oxidised metals typically have high stable emissivities, e.g. steel 0.85, copper 0.8, brass 0.7. Carbon, soot, etc. typically have an emissivity of 0.85–0.95. There is a somewhat fallible rule of thumb that anything that is dirty enough has an emissivity of 0.8. By far the most difficult target materials are clean, bright metals. Here emissivities are very low ($\leq$0.1) and may vary wildly with small amounts of surface oxidation. High-accuracy thermometry on bright metals is always demanding.

In many situations the measurement is rendered insensitive to moderate errors in emissivity setting by the choice of the shortest possible operating wavelength. For any given uncertainty in emissivity setting, the resulting uncertainty in temperature measurement is set by the inherent sensitivity of the thermometer. Table 7.2 shows this expressed in per cent change in radiance L required to produce a one degree change in indicated target temperature; the higher values at shorter wavelength clearly indicate lower errors for any given emissivity error or uncertainty.

7.3.3 Background temperature

When higher temperatures are being measured, and the effective background is much cooler than the target temperature, then the background radiance is very much lower, and the second term in equation (6) approaches zero and is frequently ignored. When measurements are made of lower temperatures, the effective background radiance is often significant in comparison with the target radiance. In this case both terms in equation (6) must be evaluated. Some thermometers allow a background temperature value $T^{\sim}$ to be 'dialled' in, but a more common practice is for $T^{\sim}$ to be taken equal to the internal temperature of the thermometer. A number of thermometer systems allow $T^{\sim}$ to be input on-line from an auxiliary sensor such as a thermocouple.

A special case exists when the surrounding temperature is very close to that of the target, and a different approach can be taken, for when $T^{\sim} = T$ equation (6) reduces to:

$$L = \{\epsilon + (1 - \epsilon)\} \cdot f(T) = f(T) \tag{7}$$

i.e., the black body condition is recovered. If the thermometer emissivity control is now set to unity, a correct temperature reading is obtained irrespective of the actual emissivity or of the background temperature adopted. Techniques exist to stimulate artificially this 'isothermal cavity' condition (see Section 7.5.1).

7.4 Practical situations

When applying infra-red thermometers the following issues must always be considered:

- Access to the target – can a clear view be obtained? Is the target in a controlled atmosphere? How big is the target? Is there any possibility of interference in the sight path, for example by smoke or steam?
- Uncertainties in the emissivity of the target – is the emissivity known? Is it high? How variable might it be?
- Contributions from the background surrounding the target – is the background hotter than the target? Is it similar to the target? Is it much colder than the target? How much does it vary?

To put the somewhat abstract discussion above into a practical context we consider three examples.

7.4.1 Hot targets in a cold background

Firstly, we consider a warm target in a background at a lower temperature, for example a food product having emerged from a cooking process. An example would be loaves of bread being air cooled on a conveyor (Fig. 7.12), where they must be below, say, 100ºC before entering a packaging machine. In this case access to the target is open and a thermometer can simply be placed above and to the side of the conveyor to obtain a clear view and avoid the hot air rising from the bread. At a distance of, say, 1.5 m from the bread a thermometer with a cone of vision as shown in Fig. 7.2 would measure a spot of 27.5 mm diameter, easily accommodated on a loaf of bread. The background is the room walls and ceiling and both are likely to be much cooler than the loaves. A thermometer should reach approximately the same temperature as the room.

Typical emissivity values for bread are not known to the author, but thought likely to be high at >0.8. Table 7.3 gives measurement errors as a function of target emissivity and various background temperatures for two thermometer operating wavelengths. The centre column applies when the emissivity 'control' is correctly set; adjacent columns apply to ±0.03 mis-settings. The calculations are based on a monochromatic thermometer at 20ºC which adopts its own temperature as the background temperature; they give a good indication of the errors expected from commercial lead sulphide (2.2 μm) and thermopile or pyroelectric (10 μm) instruments. The advantage of short wavelength operation is very apparent.

7.4.2 Hot targets in a hot background

Secondly, we consider a hot target in a background at a similar high temperature, for example a food product nearing the end of a cooking process. An example would be loaves

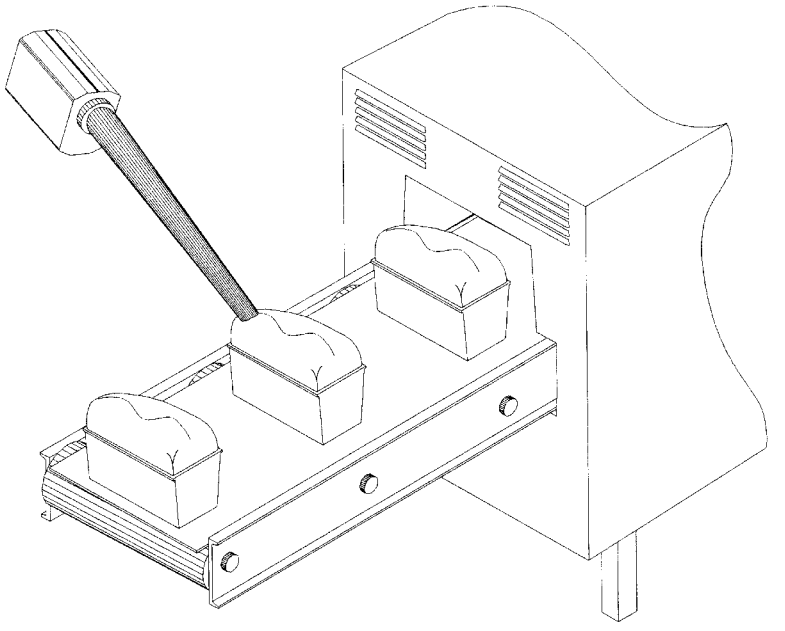


Fig. 7.12 Thermometer viewing bread on exit from oven.

of bread during baking inside an oven, where they must reach say 180ºC before leaving the oven (see Fig. 7.13). Here, access to the target is limited and a hole, possibly covered with a suitable infra-red window, will have to be provided in the oven wall in order to obtain a view of the target. At a distance of, say, 0.8 m from the bread a thermometer with a cone of vision as shown in Fig. 7.2 would measure a spot of 13 mm diameter, again easily accommodated on most loaves. The background is now the inside of the oven walls and likely to be similar to that of the loaves. In this case the thermometer, outside the oven, will be much cooler then the background surrounding the target.

Table 7.3 Measurement errors (Kelvin) for 100ºC target as a function of errors in set emissivity and assumed background temperature, for a true emissivity of 0.8.

Operating wavelength (μm)	Error in background temperature (ºC)	Emissivity ϵ		
		0.77	0.8	0.83
2.2	0	0.77	−0.04	−0.81
	10	0.78	−0.02	−0.8
	20	0.81	0	−0.77
	30	0.86	0.05	−0.73
	40	0.95	0.14	−0.64
	50	1.12	0.31	−0.47
10	0	−0.15	−2.5	−4.7
	10	1.06	−1.3	−3.5
	20	2.4	0	−2.3
	30	3.9	1.4	−0.85
	40	5.5	3	0.68
	50	7.2	4.7	2.3

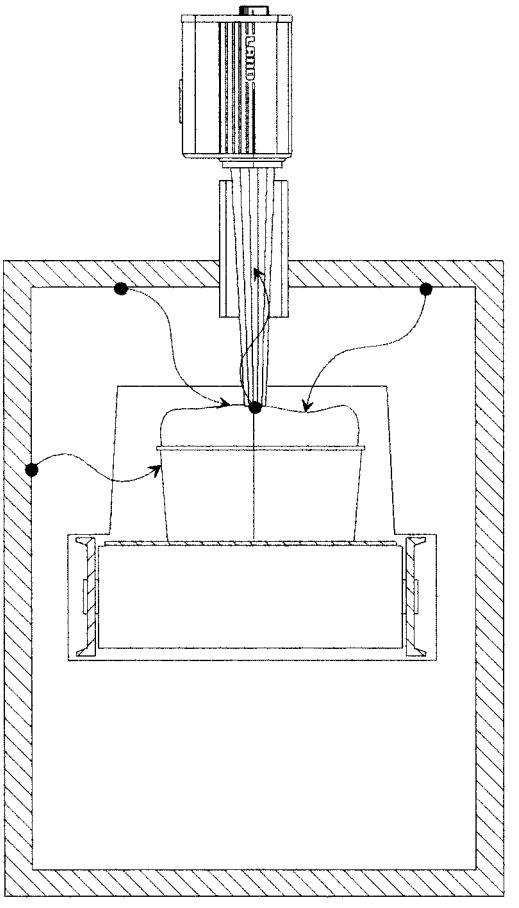


Fig. 7.13 Thermometer viewing bread inside oven.

In this near isothermal case the reflected energy is from objects at the same temperature as the target and the black body conditions are recreated as shown in equation (6). Table 7.4 gives measurement errors as a function of actual target emissivity and mismatch between background and target temperatures, for two thermometer operating wavelengths. The calculations are based on a monochromatic thermometer with its emissivity control set to 1; they give a good indication of the errors expected from commercial lead sulphide (2.2 μm) and thermopile or pyroelectric (10 μm) instruments.

The advantages of operation at the shorter, 2.2 μm, wavelength are more marginal. Considering emissivity effects, the measurement is independent of emissivity when the background and the target are exactly the same. As the background temperature falls below that of the target, then the shorter wavelength thermometer has smaller errors. When the background temperature rises above the target then the longer wavelength thermometer has the smaller errors.

7.4.3 Warm targets in a hot background

Lastly, we consider a warm target in a background at a higher temperature, for example a food product starting a cooking process. Continuing the bakery theme the example would be

Table 7.4 Measurement errors (Kelvin) for 180ºC target in near isothermal conditions as a function of background temperature and target emissivity, assuming a set emissivity of 1

Operating wavelength (μm)	Background temperature (ºC)	Emissivity ϵ 0.9	0.8	0.7
2.2	150	−2.1	−4.3	−6.6
	160	−1.6	−3.2	−4.9
	170	−0.9	−1.8	−2.7
	180	0	0	0
	190	1.1	2.2	3.3
	200	2.6	4.9	7.2
	210	4.3	8.1	11.6
10	150	−2.9	−5.8	−8.7
	160	−1.9	−3.9	−5.9
	170	−1	−2	−3
	180	0	0	0
	190	1	2	3
	200	2.1	4.1	6.1
	210	3.1	6.2	9.3

of bread dough having just entered the oven; in this case measurement may be required to prevent overheating during the early part of the bake. Again, access to the target is limited and the installation arrangement would be similar to that shown in Fig. 7.14. In this case the humidity in the oven is likely to be high and the operating wavelength should be chosen to avoid the absorption effects of the water vapour in the sight path of the thermometer. The background is again the inside walls of the oven, which at, say, 100ºC are warmer than the bread dough at, say, 60ºC. Once again, the thermometer, outside the oven, will be much cooler then the background surrounding the target. In this case the reflected energy is from objects hotter than the target and the second term in equation (5) for the reflected radiance can be larger than that emitted from the dough.

One solution for this situation is a measurement system that takes both the thermometer signal and the background temperature from, for example, a roof thermocouple, and calculates and applies a correction for the background radiation. This interpretative model assumes that the thermocouple is so positioned as to truly represent the mean background temperature ($T^{\sim}$) of the surroundings. When, as is likely in the oven, the wall and roof will be at the same temperature this positioning is trivial. However, when temperature gradients are present great care may be needed to position the thermocouple correctly.

Table 7.5 gives measurement errors as a function of emissivity and errors in measured background temperature for two thermometer operating wavelengths. Calculations are done for a true emissivity of 0.8 and the centre column applies when the emissivity 'control' is correctly set; adjacent columns apply to ±0.03 mis-settings. The calculations are again based on a monochromatic thermometer which is using a secondary input as the background temperature. The advantage of operation at the longer, 10 μm, wavelength is very apparent. Considering emissivity effects, uncertainty of between 0.77 and 0.83 result in only ±1.8ºC errors with the correct background, compared with +4, −5.5ºC for a 2.2 μm thermometer. Considering errors in the mean background value, we again see a significant benefit for the longer wavelength with less than ±4ºC error, as compared with possible out-of-range readings for a 2.2 μm thermometer. These background correcting

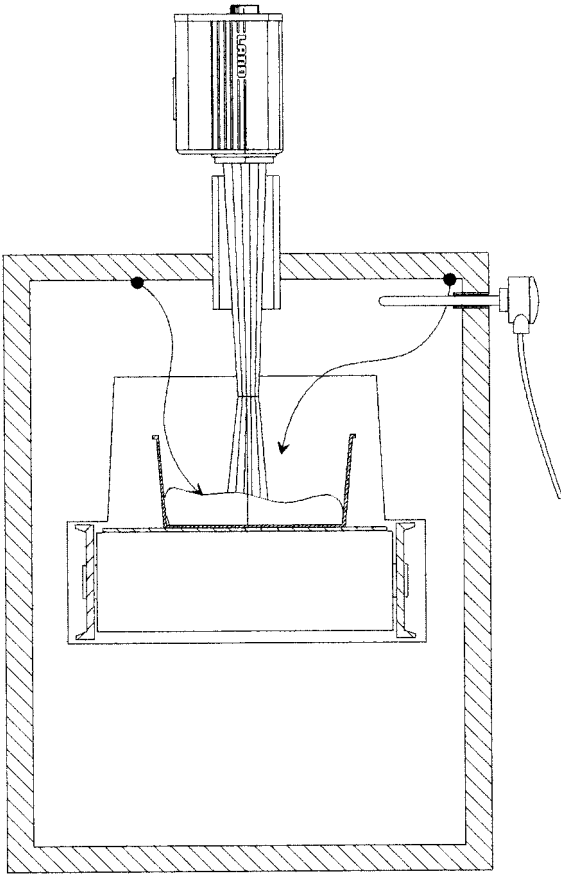


Fig. 7.14 Thermometer viewing dough inside oven.

Table 7.5 Measurement errors (kelvin) for 60ºC target in a 100ºC background as a function of errors in measured background temperature and set emissivity, for true $\epsilon = 0.8$

Operating wavelength (μm)	Error in background temperature (ºC)	Emissivity ϵ		
		0.77	0.8	0.83
2.2	−10	8.8	10.1	11.3
	−5	3.7	6.2	8.4
	−2	−1	2.9	6
	0	−5.5	0	4
	2	−12	−3.8	1.6
	5	−38	−12.5	−3.3
	10	unresolved	unresolved	−20
10	−10	1.9	3.1	4.2
	−5	0.1	1.6	2.9
	−2	−1	0.6	2.2
	0	−1.8	0	1.7
	2	−2.6	−0.7	1.1
	5	−3.8	−1.6	0.3
	10	−5.9	−3.4	−1

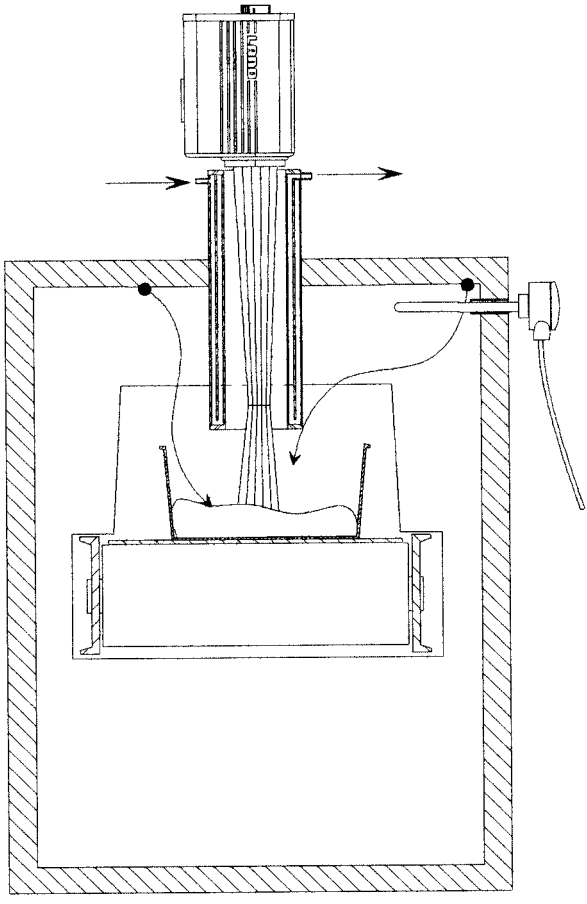


Fig. 7.15 Thermometer viewing dough through water-cooled sight tube.

systems can provide good temperature measurements when applied and installed with care, and have yielded very significant fuel savings in large furnaces.

Note: Two values for the 2.2 μm thermometer are marked unresolved, this is caused by the higher than true background measurement producing an overestimation of the correction, so that when it is applied (subtracted) the actual radiance is completely swamped giving an indication below 0K.

A second solution is artificially to create a cooler background by installing a cooled shield near the measurement point as indicated in Fig. 7.15, thereby re-creating the cold background model discussed in Section 7.4.1. The approach has been widely and successfully used in situations where continuous strip or sheet materials are being processed.

7.5 Miscellaneous techniques

In this section we look firstly at simple means of enhancing emissivity, then at some techniques which avoid the need to know emissivity in specific conditions, and finally at ways of measuring inaccessible targets.

7.5.1 Enhancing emissivity

A simple way to improve the emissivity of, say, a bright metal target is to paint it. Paint with high (~0.9) stable emissivities are available, permitting accurate temperature measurement in virtually any background conditions. Another, though somewhat restricted, approach is to create an isothermal cavity within the body to be measured, for example by drilling a hole. A hole six times deeper than its diameter, will produce an effective emissivity close to unity, especially if the walls are roughened and the bottom is conical rather than square to the axis. Cracks, grooves or existing cavities on the target surface can also achieve similar effects. These can be particularly useful in measuring products on a conveyor belt, when a small target spot is used and the very fast responding thermometer output signal is peak picked to measure the high radiance values from these high emissivity features.

The effect of the isothermal cavity can be simulated by surrounding the target spot with a high reflectance enclosure, the background now becomes a reflection of the target and appears to be at the target temperature thereby re-creating the black body conditions of equation (6). The first product to use this technique was designed in the 1950s and comprised a gold (highly reflective in the infra-red) plated hemisphere with a small hole through which the measurement is made. The 'gold cup' was placed within the periphery of the hemisphere on the target surface and temperature measurements virtually independent of the surface emissivity were possible. The exclusion of background radiation also simplified the mathematical model used for interpretation of results. This highly successful product lasted many years but in requiring surface contact, it lost the inherent non-contact feature of infra-red thermometry, and made it ever more unsuitable for modern continuous processes.

Several similar approaches have been used where the reflector is fixed close to a moving target to form a wedge-shaped cavity with the target surface. One creative approach for strip or sheet products is to use the naturally occurring wedges formed where target material spools off a shiny reflective roller. This has been proven to provide very reliable measurements even when target emissivity is quite low. An interesting variation is created by allowing the reflector temperature to be close to the target, in this case the precise reflectance value of the reflector becomes non-critical as isothermal conditions are recreated. Recently, the spherical geometry of the gold cup has been reviewed with the aim of removing the requirement to contact the surface, and new designs allow clearance gaps of approximately half the reflector diameter whilst maintaining high degrees of emissivity enhancement on many low-emissivity target surfaces.

7.5.2 Avoiding the need to know emissivity

The Holy Grail of infra-red thermometry will always be a truly emissivity independent measurement technique. Over the 50+ years since the first practical infra-red thermometers were built, a number of techniques have been used which claim to have found this Holy Grail. In truth they avoid the need for known emissivity only at the expense of some other assumption about the target surface. They have value, but tend to work only in very specific conditions. Unfortunately, the general image of infra-red thermometry has been tarnished by overly aggressive marketing of some of the following methods.

A definitive instance is ratio (two-colour) thermometry. Here radiance is sensed in two separate wavebands and a signal is derived which is proportional to the ratio of the

respective energies. The argument is that, while the energy in each waveband depends on emissivity, this cancels out the ratio, rendering the ratio signal a function only of temperature. This is true only for a grey target, whose emissivity is independent of wavelength. In general the ratio signal depends on the ratio of the respective emissivities in the two wavebands. Moreover, a ratio thermometer is typically several times more sensitive to a mis-setting of emissivity ratio than is a conventional thermometer to a mis-setting of emissivity. Ratio thermometers are actually more useful for coping with a partially obstructed or unfilled field of view, the ratio signal being unaffected as long as attenuation is identical in both wavebands.

Certain two-colour thermometers derive calibrated radiances in two wavebands in addition to the ratio. This additional information can, in certain very specific circumstances, be used to deduce emissivity on-line. A variety of reflection methods exist which derive emissivity on-line. Looking back to equation (5), if the background temperature $T^{\sim}$ can be oscillated between two known values then the value for $1-\epsilon$ and hence emissivity can be deduced from the a.c. ripple on the thermometer signal. This is practicable only for highly specular targets, where a reasonably small variable temperature source at the specular angle suffices.

If the background source is varied slowly and a polarising filter is rotated rapidly in front of the thermometer, an a.c. ripple is produced (due to variation of emissivity with polarisation) which falls to zero when the background temperature equals the target temperature. Simultaneous monitoring of background source temperature and thermometer output ripple thus permits an emissivity-free measurement. The requirement for high specularity is eased if one can assume that non-specular reflection is unpolarised.

Recently, thermometers have been promoted which incorporate a laser reflectometer. A laser beam illuminates the target, and the fraction of beam power reflected back into the instrument is monitored. If this can be related to the total beam power reflected – i.e. over all angles – then emissivity is derivable from the relationship: emissivity equals one minus total reflectance. Clearly this entails a model relating reflectance into the solid angle subtended by the instrument's collecting aperture to total (i.e. 2π solid angle) reflectance – a point not overemphasised by some vendors of the equipment. In practice the assumption made is pure diffuse reflection. The scope of validity of this assumption is not yet well documented. A two-wavelength version allows the emissivity ratio to be derived given only that the collected to total reflectance ratio is the same at both wavelengths.

7.5.3 Inaccessible targets

As discussed earlier, the measurement of targets whilst in the middle of processes is often required in order to provide the best parameters for control of the process. If the product is a liquid flowing in a pipe, then measurement by thermocouple fitted through the tube wall, can give accurate steady state data. However, it may take several seconds for the protective sheath and hence the sensor within to respond to process changes. As soon at the product is a solid or individual product item then non-contact infra-red thermometry is needed. To allow the instrument to access the target within the process, infra-red transmitting windows may be used to provide a viewing port to the product. With suitable materials and design, very high pressures (~200 atmospheres) can be contained by these windows, even permitting measurements in extruders.

The majority of window materials conventionally used are of doubtful applicability in food processes on grounds of toxicity, solubility or propensity to shatter into sharp

fragments. Probably the most suitable material, by a considerable margin, is synthetic sapphire (single-crystal alumina). The relevant properties of sapphire are given in Table 7.5. It is a relatively inexpensive material in window diameters up to 25 mm or so. Fixture is generally via a suitable gasket or O-ring, but sapphire can also be brazed to Nilo or titanium. Sapphire is highly transparent from the visible out to wavelengths of about 5μm, restricting its use as a window to shorter wavelength thermometer types. Windowing the longer wavelength, 8 to 14 μm, types is much more problematical. Recent developments have allowed the measurement of temperatures below freezing point with 3 to 5 μm thermometers, thereby extending the range of processes measurable through a sapphire window, see Table 7.1 for the corresponding temperature ranges. The usual materials – germanium, zinc sulphide, barium fluoride – are very doubtful for the reasons given above. Polyethylene transmits reasonably well, but only in thin films (0.1 mm).

Targets that are inaccessible due to high ambient temperature, vibration or radio frequency (RF) fields, can sometimes be measured using fibre optics. The infra-red collection assembly is separated from the detector unit with a flexible optic fibres. This allows the sensitive detector unit to be fixed in a benign environment. Collection assemblies can be built very tough, for example to withstand ambient temperatures over 400ºC. Figure 7.6 on page 195 shows a typical instrument. The use of fibre optics, with commonly available glass and silica fibre, is limited to moderately short wavelength thermometers, owing to the transmission characteristics of the fibre. Once again recent



Fig. 7.16 Vanguard thermometer for chilled packaged foods.

developments have allowed measurements as low as 50ºC with a 2.2μm instrument using quartz fibre bundles. Typically, fibre lengths of a few metres are supplied, but much longer lengths are technically possible.

A product combining several of these features has recently been developed to measure the temperature of finished food items within their packaging (Fig. 7.16), without destroying the product. This uses a gold cup type reflector to minimise the effects of package emissivity variations. It still measures the package outer surface and therefore has been arranged so as to view upwards to the bottom surface of the target so that gravity keeps the content and package in contact. It also utilises a sapphire window to protect the reflector, made possible by a moderately short wavelength thermometer. Trials in food distribution have shown excellent correlation to the traditional destructive method of inserting a thermocouple type probe through the package.

7.6 Further reading

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